PROJECT REPORT

DESIGN AND DEVELOPMENT OF A SIMPLE CALCULATOR IN C

Arithmetic operations using user-defined functions and control structures

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1. Abstract

This project presents the design and development of a simple menu-driven calculator using the C programming language. The program performs four basic arithmetic operations: addition, subtraction, multiplication, and division. Each operation is implemented through a separate user-defined function. A **switch-case** statement is used to select the operation, while an **if-else** condition prevents division by zero. The project demonstrates modular programming, function calls, decision-making statements, input/output operations, and basic error handling.

2. Introduction

A calculator is a common application that carries out mathematical calculations. Building a calculator in C is a useful beginner-level programming exercise because it combines several fundamental concepts in one program. In this project, the user enters two numbers and selects an arithmetic operation from a menu. The program then calls the relevant function and displays the result.

The program is designed to be straightforward, readable, and easy to modify. Separating each arithmetic operation into its own function makes the code more organised and supports code reuse.

3. Objectives

• To develop a menu-driven calculator using the C programming language.

• To perform addition, subtraction, multiplication, and division.

• To understand how to declare, define, and call user-defined functions.

• To apply control structures such as switch-case and if-else.

• To validate the division operation and handle division by zero.

• To display clear results and messages for invalid menu choices.

4. Software and Hardware Requirements

Category	Requirement
Programming language	C
Compiler	GCC, Clang, or another standard C compiler
Editor / IDE	VS Code, Code::Blocks, Dev-C++, or any text editor
Hardware	A computer or laptop
Operating system	Windows, macOS, or Linux

5. Concepts Used

5.1 User-defined functions: Functions are reusable blocks of code designed to perform a specific task. This program uses four functions: **add()**, **subtract()**, **multiply()**, and **divide()**. Each function receives two numbers and returns the calculated value.

5.2 switch-case statement: The switch statement checks the menu choice and transfers control to the matching case. The **break** statement prevents execution from continuing into the next case.

5.3 if-else statement: Before division, the program checks whether the second number is zero. Division is performed only when the divisor is not zero.

5.4 Input and output: The **printf()** function displays messages and results, while **scanf()** accepts input from the user.

6. Algorithm

1. Start the program.
2. Declare two floating-point variables for the numbers, a result variable, and an integer for the menu choice.
3. Ask the user to enter the first and second numbers.
4. Display the four arithmetic operations and ask the user to enter a choice.
5. Use switch-case to identify the selected operation.
6. Call the corresponding user-defined function and calculate the result.
7. For division, first check whether the second number is zero. If it is zero, display an error message.
8. Display the result or an appropriate error message.
9. End the program.

7. Flow of the Program

Step	Action
1	Start
2	Input two numbers
3	Display operation menu
4	Read user's choice
5	switch-case selects an operation
6	Call the matching function
7	For division, check divisor is not zero
8	Display result or error
9	Stop

Textual flow: Start → Input numbers → Show menu → Read choice → Select operation → Call function → Check division condition if required → Display output → Stop.

8. Source Code

```
#include <stdio.h>

// Function declarations
float add(float a, float b);
float subtract(float a, float b);
float multiply(float a, float b);
float divide(float a, float b);

int main()
{
    float num1, num2, result;
    int choice;

    printf("===== SIMPLE CALCULATOR =====\n");

    printf("Enter first number: ");
    scanf("%f", &num1);
```

```
    printf("Enter second number: ");
    scanf("%f", &num2);

    printf("\nSelect an operation:\n");
    printf("1. Addition (+)\n");
    printf("2. Subtraction (-)\n");
    printf("3. Multiplication (*)\n");
    printf("4. Division (/)\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            result = add(num1, num2);
            printf("Result = %.2f\n", result);
            break;

        case 2:
            result = subtract(num1, num2);
            printf("Result = %.2f\n", result);
            break;

        case 3:
            result = multiply(num1, num2);
            printf("Result = %.2f\n", result);
            break;

        case 4:
            if (num2 != 0)
            {
                result = divide(num1, num2);
                printf("Result = %.2f\n", result);
            }
            else
            {
                printf("Error: Division by zero is not allowed.\n");
            }
            break;

        default:
            printf("Invalid choice!\n");
    }

    return 0;
}

// Function definitions
float add(float a, float b)
{
    return a + b;
}

float subtract(float a, float b)
{
    return a - b;
}

float multiply(float a, float b)
{
    return a * b;
}

float divide(float a, float b)
{
    return a / b;
}
```

9. Explanation of the Source Code

Header file: #include <stdio.h> provides standard input and output functions such as printf() and scanf().

Function declarations: The four declarations inform the compiler about the function names, return types, and parameters before they are called in main().

main() function: This is the starting point of program execution. It accepts two numbers and the user's menu choice.

switch-case: It directs the program to the selected arithmetic operation. Each case calls one function, prints the result, and uses break to exit that case.

Division validation: The if-else condition checks num2 != 0. This avoids attempting to divide by zero.

Function definitions: Each function performs one calculation and returns its answer to main(). The %.2f format specifier displays the result rounded to two decimal places.

10. Sample Outputs

Example 1 – Addition

```
Enter first number: 20
Enter second number: 10
Enter your choice: 1
Result = 30.00
```

Example 2 – Subtraction

```
Enter first number: 20
Enter second number: 10
Enter your choice: 2
Result = 10.00
```

Example 3 – Multiplication

```
Enter first number: 20
Enter second number: 10
Enter your choice: 3
Result = 200.00
```

Example 4 – Division

```
Enter first number: 20
Enter second number: 10
Enter your choice: 4
Result = 2.00
```

Example 5 – Division by zero

```
Enter first number: 20
Enter second number: 0
Enter your choice: 4
Error: Division by zero is not allowed.
```

11. Test Cases

Test	Input (a, b)	Choice	Expected output
Addition	12, 8	1	20.00
Subtraction	12, 8	2	4.00
Multiplication	12, 8	3	96.00
Division	12, 4	4	3.00
Division by zero	12, 0	4	Error message
Invalid choice	12, 8	9	Invalid choice message

12. Advantages

• The program is simple and easy to understand.

• Functions divide the program into smaller, reusable parts.

• The switch-case structure makes operation selection clear.

• The division-by-zero check prevents an invalid arithmetic operation.

• The program can be extended with additional operations.

13. Limitations

• The program performs only one operation each time it runs.

• It supports four basic arithmetic operations only.

• It does not validate non-numeric input entered by the user.

• It does not include advanced operations such as modulus, powers, or square roots.

14. Possible Future Enhancements

The calculator can be improved by adding a loop so that users can perform multiple calculations without restarting the program. Additional features may include modulus, percentage, power, square root, and improved input validation. A graphical user interface could also be developed as an extension.

15. Conclusion

The project successfully demonstrates how to create a basic calculator in C using user-defined functions and control structures. The program performs addition, subtraction, multiplication, and division through separate functions. The switch-case statement manages the user's selection, while if-else handles the special case of division by zero. Overall, this project strengthens understanding of functions, decision-making, and structured programming in C.

16. References

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